



ODRE PQC

PUBLIC RELEASE EVIDENCE

Release Evidence

Security Validation Across Windows Server 2022 and Ubuntu 24.04 LTS

Evidence date: 2026-09-10 KST

**PASS WITH DOCUMENTED
SCOPE**

Documented scope

Release Gates 1 through 4 were completed within their documented scope on isolated Windows Server 2022 AMD64 and Ubuntu 24.04 LTS AMD64 virtual machines. The recorded evidence covers security attacks, replay and tamper rejection, state and Core failures, resource pressure, restart and recovery, bounded concurrency, and one-hour protected soak testing.

The tested protected path was the server-side Gateway to ODRE v3 Core to Protected Handler boundary. Client-to-server traffic remains HTTPS/TLS. This report does not claim mobile-to-server end-to-end PQC.

0

Protected
Handler bypass

0

Handler double
execution

0

Plaintext fallback

0

Classical fallback



Validated platforms

Platform	Result	Scope
Windows Server 2022 AMD64	PASS	Frozen Windows candidate; bounded safe resource matrix
Ubuntu 24.04 LTS AMD64	PASS	Frozen Ubuntu candidate; bounded safe resource matrix

Validation coverage

- Replay and tamper rejection
- Invalid and expired session handling
- Protected Handler bypass testing
- Core unavailability
- State read and write failure
- Session expiry, restart, and recovery

Security under load

Each operating system passed 16 of 16 security and error cases. In the mixed workload, 20 of 20 legitimate requests completed and 4 of 4 attack requests were rejected on each operating system. Attack requests executed the Protected Handler zero times.

Observed within the documented validation scope

- CPU, memory, FD, and handle pressure
- Bounded concurrency
- One-hour protected soak testing
- Duplicate Handler execution checks
- Plaintext and classical fallback checks

Bounded concurrency observations

Intentional GATEWAY_BUSY admission responses are shown separately from unexpected errors. These measurements are not a supported-capacity guarantee.

OS	Concurrency	Issued	Success	Bounded 503	Other
Windows	50	50	16	34	0
Windows	100	100	17	83	0
Ubuntu	50	50	16	34	0
Ubuntu	100	100	17	83	0



One-hour protected soak observations

The latency distribution covers the recorded protected soak configuration. Resource values were for the entire harness-owned process tree, not product-only consumption.

OS	Duration	Requests	Errors	p50	p95	p99
Windows	3,600.859 s	3,597	0	10.531 ms	25.808 ms	33.950 ms
Ubuntu	3,600.846 s	3,587	0	114.289 ms	208.049 ms	334.666 ms

Resource-pressure and recovery

Windows Server 2022	Ubuntu 24.04 LTS
- CPU: Job CPU hard cap 25 percent with affinity to one logical CPU; 20 of 20 protected requests succeeded. - Memory: JobMemoryLimit 234,881,024 bytes; the controlled allocation reached MemoryError and service recovered to HTTP 200. - Handles: 2,048 live handles plus injected outbound allocation failure produced fail-closed HTTP 503 with Handler count 0; recovery returned HTTP 200 with one Handler execution.	- CPU: process affinity to one logical CPU; 20 of 20 protected requests succeeded. - Memory: RLIMIT_AS 751,669,248 bytes; the controlled allocation reached MemoryError and service recovered to HTTP 200. - FD: RLIMIT_NOFILE soft limit 64; errno 24 was observed at the real process limit. Protected requests failed closed with Handler count 0 and recovered to HTTP 200.

Important Windows handle limitation	Windows does not provide a Linux RLIMIT_NOFILE-equivalent safe per-process kernel handle-count quota. The test retained 2,048 real handles and injected an outbound-allocation failure; it did not exhaust the operating system kernel handle limit. Ubuntu did reach the configured RLIMIT_NOFILE limit.
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Artifact identity and test isolation

Frozen candidate manifest SHA-256	FBF2C2F453203797A9D47E6269B9B036F3455B3D7C73AE36E5ADA7DAA6E28402
Sealed Core v0.2.9 SHA-256	A8AFA10EE0AFACEF1C0FAF55FF07B96C722920E5AEE8692F6F026E804F028697
Windows Native SHA-256	12E9D3E8B702DB4447B50B95D089790A21E154791F68A9095EF9C21334D41B2C
Ubuntu Native SHA-256	08B3C42A0F62B2EF2D3128372560BD7020278300EA3EB52DE645388F7E7143E3

Product modification: 0 | Core modification: 0 | Production contact: 0 | Real payment/mail/customer Unit change: 0

How to interpret this evidence

Supported statement

Within the recorded isolated configurations, tested security, state, and resource failures did not bypass the Protected Handler and the path failed closed. Recorded bounded concurrency and one-hour soak results are reproducible evidence for that documented configuration.

Do not infer

- A maximum processing-capacity or 100-concurrent-client guarantee.
- The same RPS or latency on every hardware, network, or customer workload.
- Proof that a long-term memory leak is impossible.
- Windows kernel handle-limit exhaustion.
- Coverage of WAF duties, authenticated malicious business input, or complete root or OS compromise.

Release evidence conclusion

Windows Server 2022: PASS. Ubuntu 24.04 LTS: PASS. ODRE PQC release evidence: PASS WITH DOCUMENTED SCOPE.

These results represent documented ODRE PQC release evidence. They do not constitute an independent third-party certification or external security accreditation. This evidence does not by itself represent complete customer Production release approval.

Prepared from the frozen Gate 4 final evidence report. Evidence date: 2026-09-10 KST.